Agile Development Process and Devops

AGILE LAB FAT

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Register Number : 23MIS0634

Slot: C1+TC1

SET – 2

ANSWER ALL THE QUESTIONS

Online Auction System

Problem Statement:

Ensure bids are recorded and highest bidder is selected correctly.

Tasks:

1. Develop bidding logic

2. Write JUnit tests for bid validation

3. Implement CI/CD pipeline with automated testing

➤ Set up version control using GitHub

➤ Configure Cl pipeline (e.g., Jenkins/GitHub Actions) to trigger on code commits

➤ Automate build process using Maven

➤ Integrate automated unit tests in pipeline

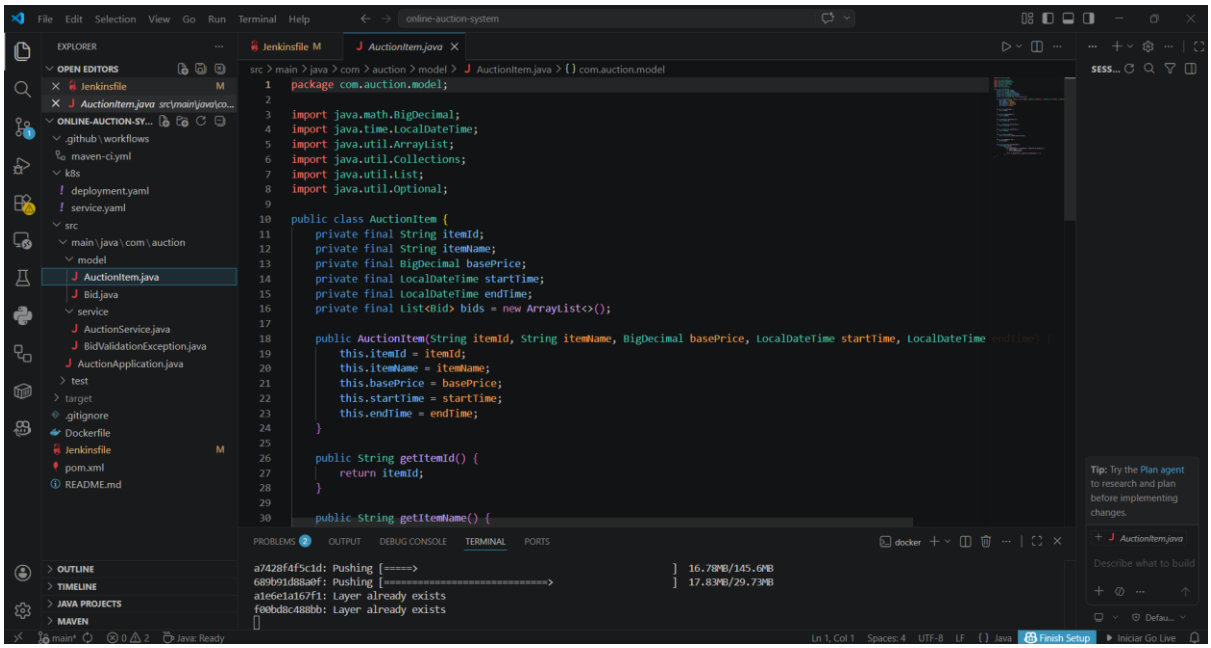
➤ Configure CD pipeline for automatic deployment to staging/production

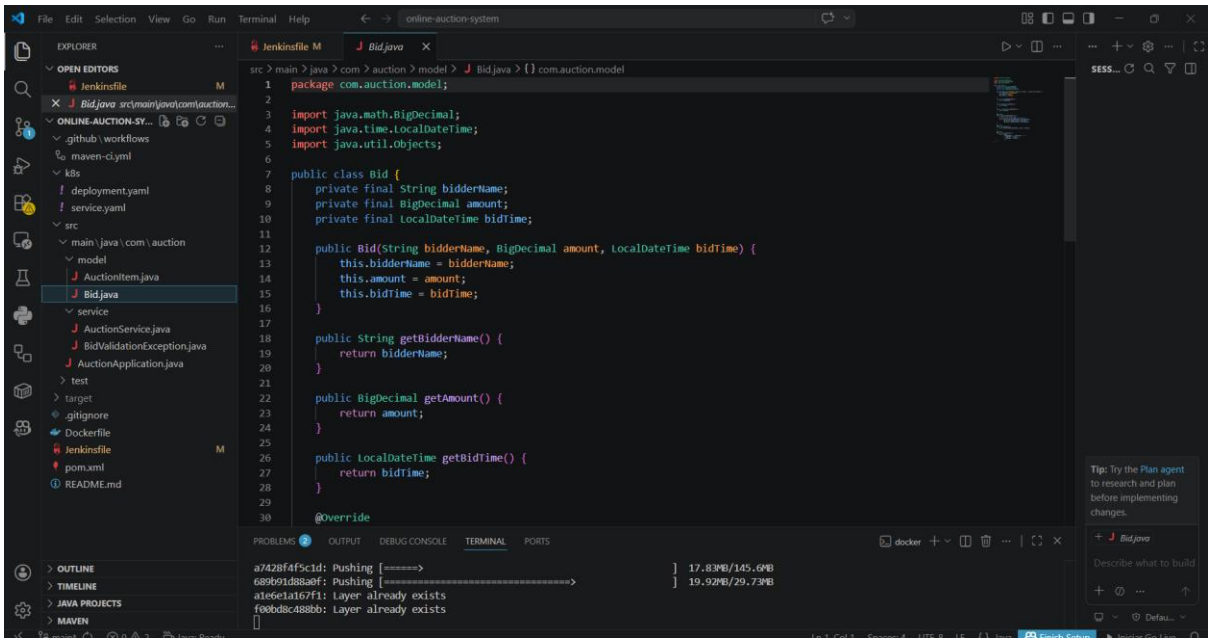
4. Deploy services on Docker and Kubernetes

➤ Containerize application using Docker (write Dockerfile)

➤ Build and push Docker image to container registry (Docker Hub)

➤ Create Kubernetes deployment and service YAML files





mvn clean package:

```
Downloading from central: Follow link (ctrl + click) apache.org/maven2/org/vafer/jdependency/2.10/jdependency-2.10.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/ow2/asm/asm-commons/9.7/asm-commons-9.7.jar (73 kB at 198 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/ow2/asm/asm-tree/9.7/asm-tree-9.7.jar (52 kB at 88 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/ow2/asm/asm/9.7/asm-9.7.jar (125 kB at 199 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/vafer/jdependency/2.10/jdependency-2.10.jar (416 kB at 480 kB/s)
[INFO] Replacing original artifact with shaded artifact.
[INFO] Replacing C:\Users\rakes\Downloads\online-auction-system\target\online-auction-system-1.0.0.jar with C:\Users\rakes\Downloads\online-auc
tion-system\target\online-auction-system-1.0.0-shaded.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  10.063 s
[INFO] Finished at: 2026-04-15T18:01:05+05:30
[INFO] ------------------------------------------------------------------------
PS C:\Users\rakes\Downloads\online-auction-system>
```

mvn clean test:

```
PROBLEMS 2   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS        powershell

[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.auction.service.AuctionServiceTest
[INFO] Tests run: 7, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.087 s -- in com.auction.service.AuctionServiceTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 7, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  3.528 s
[INFO] Finished at: 2026-04-15T18:02:01+05:30
[INFO] ------------------------------------------------------------------------
PS C:\Users\rakes\Downloads\online-auction-system>
```

git init

git add.

```
PS C:\Users\rakes\Downloads\online-auction-system> git init
Initialized empty Git repository in C:/Users/rakes/Downloads/online-auction-system/.git/
PS C:\Users\rakes\Downloads\online-auction-system> git add .
warning: in the working copy of '.github/workflows/maven-ci.yml', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of '.gitignore', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'Dockerfile', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'Jenkinsfile', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'README.md', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'k8s/deployment.yaml', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'k8s/service.yaml', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'pom.xml', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'src/main/java/com/auction/AuctionApplication.java', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'src/main/java/com/auction/model/AuctionItem.java', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'src/main/java/com/auction/model/Bid.java', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'src/main/java/com/auction/service/AuctionService.java', LF will be replaced by CRLF the next time Git touches
it
warning: in the working copy of 'src/main/java/com/auction/service/BidValidationException.java', LF will be replaced by CRLF the next time Git
```

```
PS C:\Users\rakes\Downloads\online-auction-system> git add .
warning: in the working copy of '.github/workflows/maven-ci.yml', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of '.gitignore', LF will be replaced by CRLF the next time Git touches it
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warning: in the working copy of 'k8s/deployment.yaml', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'k8s/service.yaml', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'pom.xml', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'src/main/java/com/auction/AuctionApplication.java', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'src/main/java/com/auction/model/AuctionItem.java', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'src/main/java/com/auction/model/Bid.java', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'src/main/java/com/auction/service/AuctionService.java', LF will be replaced by CRLF the next time Git touches
it
warning: in the working copy of 'src/main/java/com/auction/service/BidValidationException.java', LF will be replaced by CRLF the next time Git
touches it
warning: in the working copy of 'src/test/java/com/auction/service/AuctionServiceTest.java', LF will be replaced by CRLF the next time Git touc
```

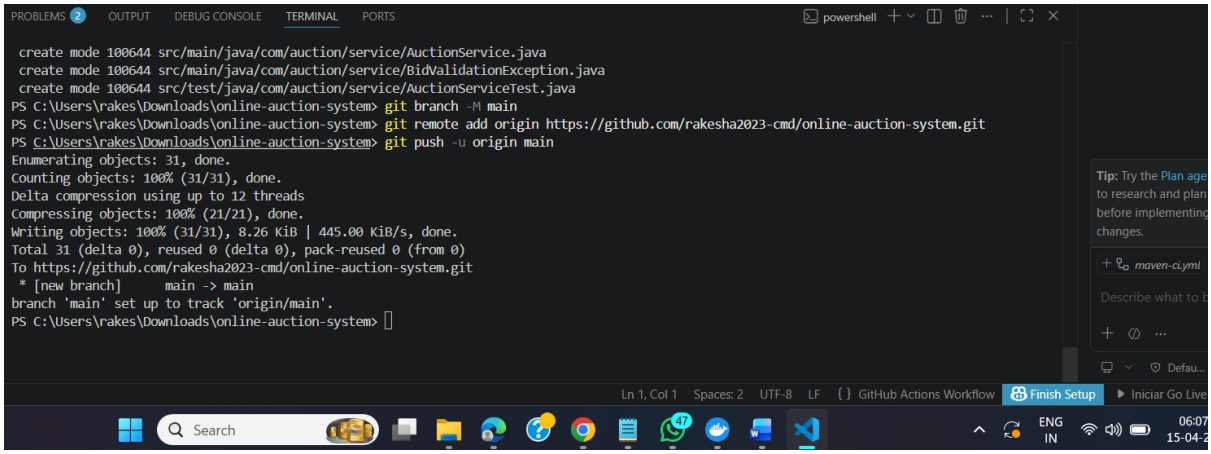
git commit -m "first commit":

```
hes it
PS C:\Users\rakes\Downloads\online-auction-system> git commit -m "first commit"
[master (root-commit) 2f061fa] first commit
 14 files changed, 757 insertions(+)
 create mode 100644 .github/workflows/maven-ci.yml
 create mode 100644 .gitignore
 create mode 100644 Dockerfile
 create mode 100644 Jenkinsfile
 create mode 100644 README.md
 create mode 100644 k8s/deployment.yaml
 create mode 100644 k8s/service.yaml
 create mode 100644 pom.xml
 create mode 100644 src/main/java/com/auction/AuctionApplication.java
 create mode 100644 src/main/java/com/auction/model/AuctionItem.java
 create mode 100644 src/main/java/com/auction/model/Bid.java
 create mode 100644 src/main/java/com/auction/service/AuctionService.java
 create mode 100644 src/main/java/com/auction/service/BidValidationException.java
 create mode 100644 src/test/java/com/auction/service/AuctionServiceTest.java
PS C:\Users\rakes\Downloads\online-auction-system> git branch -M main
```

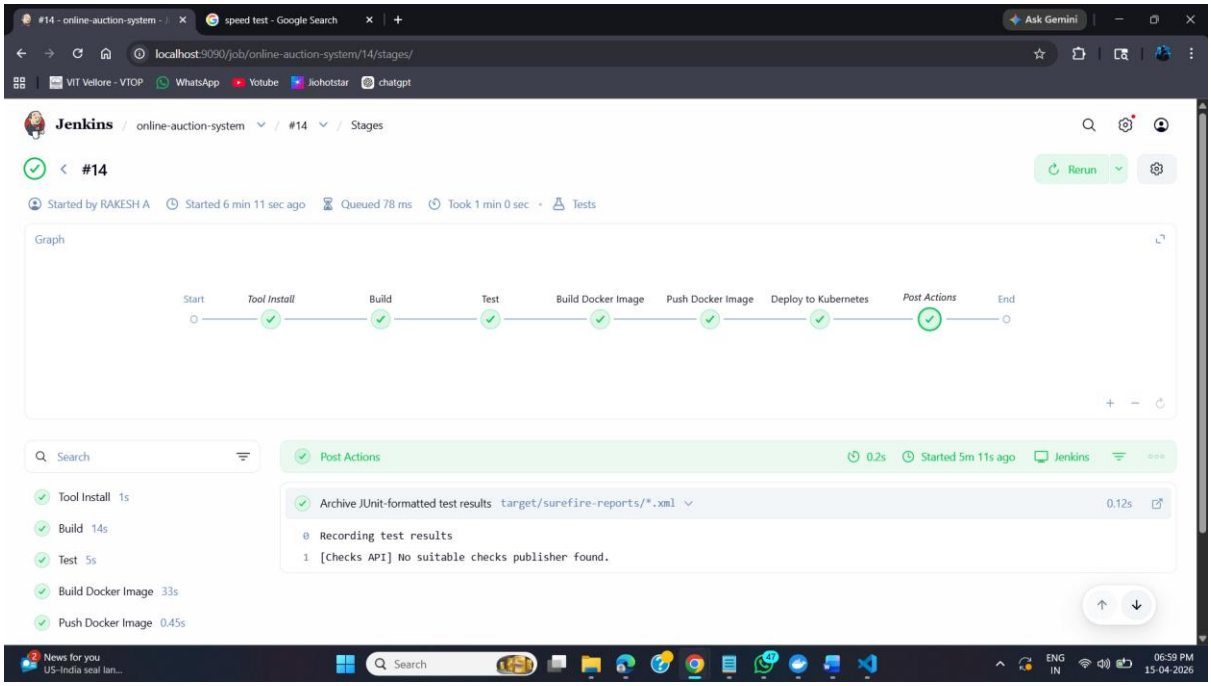
git branch -M main

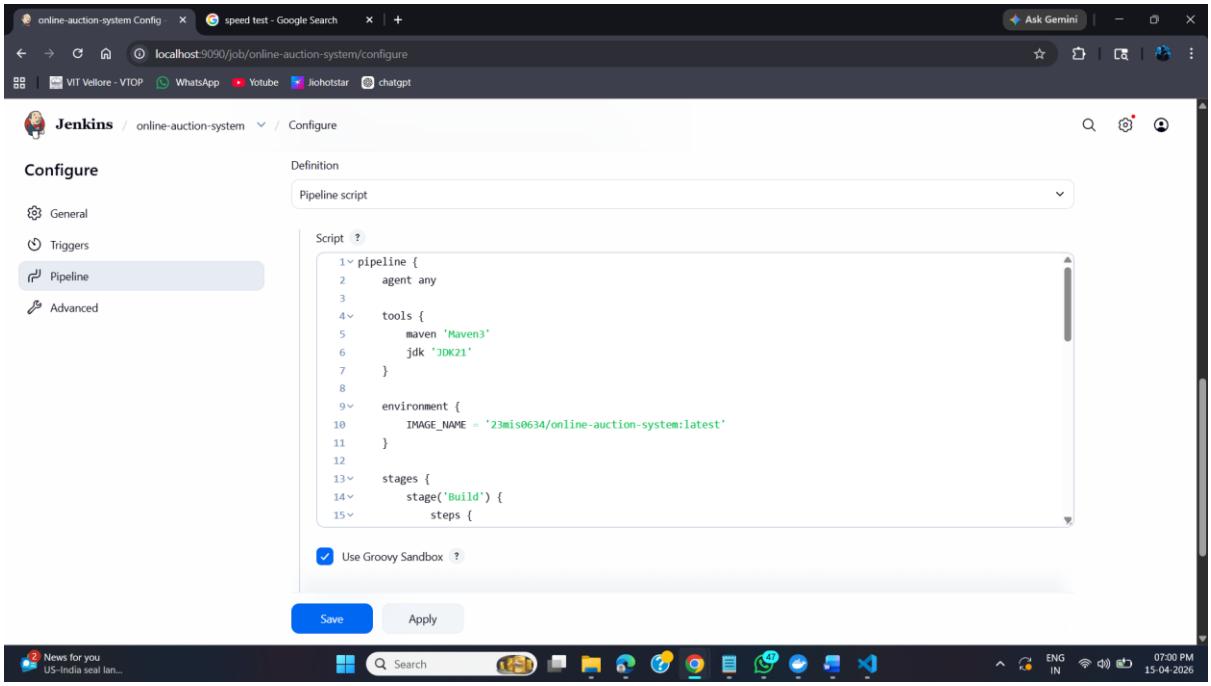
git remote add origin https://github.com/rakesha2023-cmd/online-auction-system.git

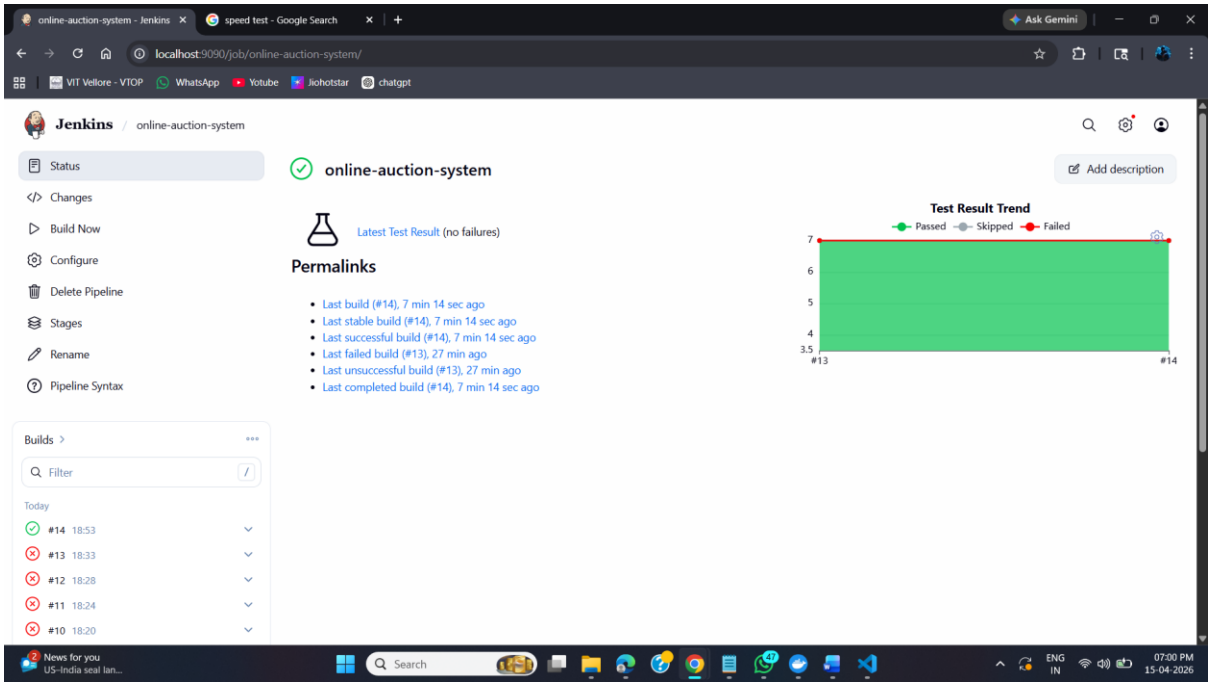
git push -u origin main



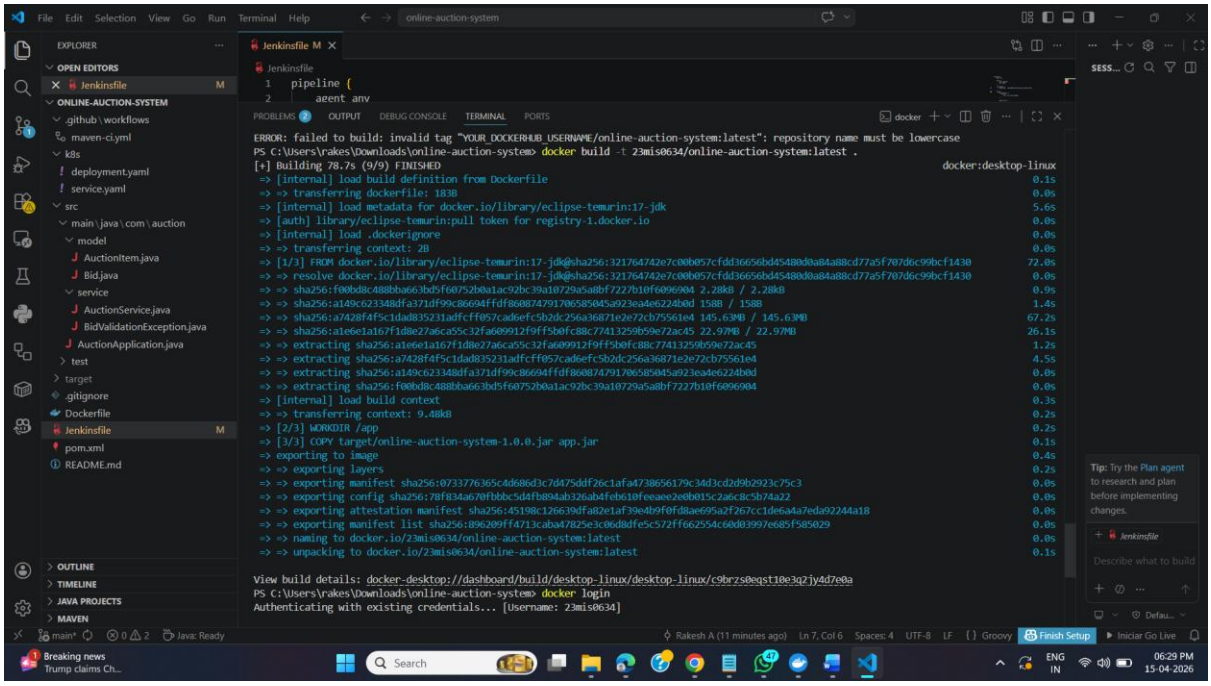
Jenkins:




online-auction-system Config
speed test - Google Search
localhost:9090/job/online-auction-system/configure
Jenkins / online-auction-system / Configure
Configure
General
Triggers
Pipeline
Advanced
Definition
Pipeline script
Script
pipeline {
agent any
tools {
maven 'Maven3'
jdk 'JDK21'
}
environment {
IMAGE_NAME = '23mis0634/online-auction-system:latest'
}
stages {
stage('Build') {
steps {
Use Groovy Sandbox
Save
Apply


online-auction-system - Jenkins
speed test - Google Search
localhost:9090/job/online-auction-system/
Jenkins / online-auction-system
Status
Changes
Build Now
Configure
Delete Pipeline
Stages
Rename
Pipeline Syntax
online-auction-system
Add description
Latest Test Result (no failures)
Permalinks
Last build (#14), 7 min 14 sec ago
Last stable build (#14), 7 min 14 sec ago
Last successful build (#14), 7 min 14 sec ago
Last failed build (#13), 27 min ago
Last unsuccessful build (#13), 27 min ago
Last completed build (#14), 7 min 14 sec ago
Test Result Trend
Passed
Skipped
Failed
Builds
Filter
Today
#14 18:53
#13 18:33
#12 18:28
#11 18:24
#10 18:20

docker build -t 23mis0634/online-auction-system:latest .

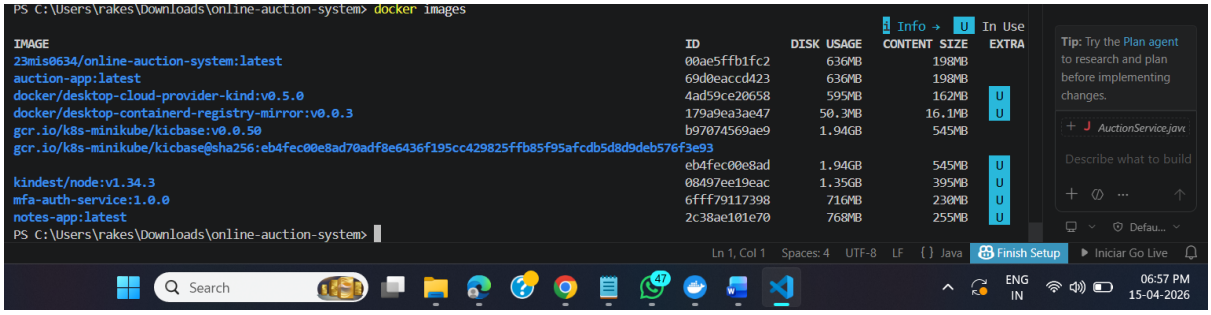


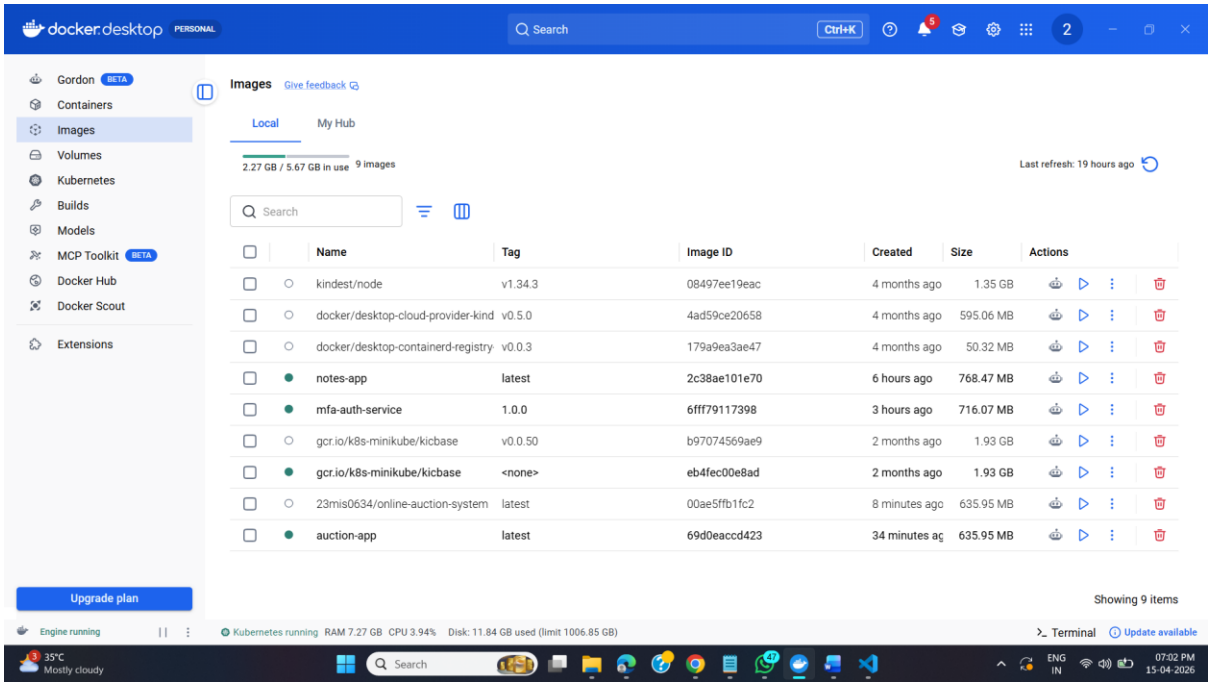
docker login:

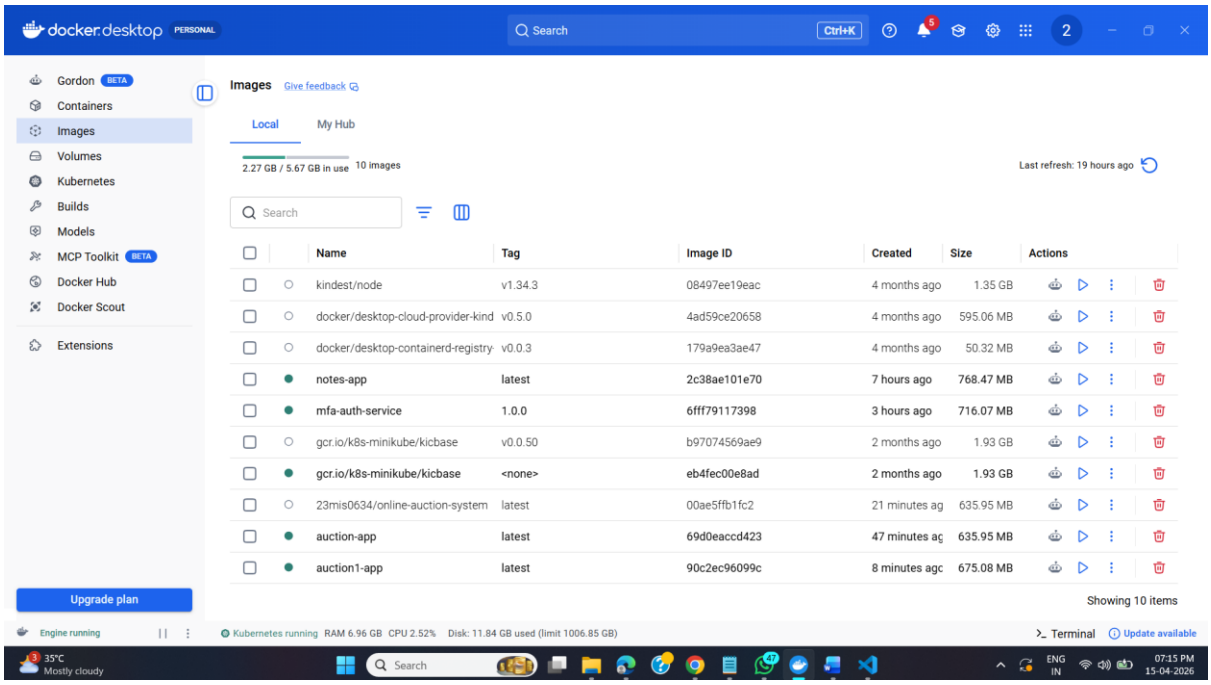
```
View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/c9brzs0eqst10e3q2jy4d7e0a
PS C:\Users\rakes\Downloads\online-auction-system> docker login
Authenticating with existing credentials... [Username: 23mis0634]

i Info → To login with a different account, run 'docker logout' followed by 'docker login'


Login Succeeded
```




docker desktop PERSONAL
Search Ctrl+K
Gordon BETA
Containers
Images
Volumes
Kubernetes
Builds
Models
MCP Toolkit BETA
Docker Hub
Docker Scout
Extensions
Upgrade plan
Images Give feedback
Local My Hub
2.27 GB / 5.67 GB in use 9 images
Last refresh: 19 hours ago
Search
Name Tag Image ID Created Size Actions
kindest/node v1.34.3 08497ee19eac 4 months ago 1.35 GB
docker/desktop-cloud-provider-kind v0.5.0 4ad59ce20658 4 months ago 595.06 MB
docker/desktop-containerd-registry v0.0.3 179a9ea3ae47 4 months ago 50.32 MB
notes-app latest 2c38ae101e70 6 hours ago 768.47 MB
mfa-auth-service 1.0.0 6fff79117398 3 hours ago 716.07 MB
gcr.io/k8s-minikube/kicbase v0.0.50 b97074569ae9 2 months ago 1.93 GB
gcr.io/k8s-minikube/kicbase <none> eb4fec00e8ad 2 months ago 1.93 GB
23mis0634/online-auction-system latest 00ae5ffb1fc2 8 minutes ago 635.95 MB
auction-app latest 69d0eaccd423 34 minutes ag 635.95 MB
Showing 9 items
Engine running
Kubernetes running RAM 7.27 GB CPU 3.94% Disk: 11.84 GB used (limit 1006.85 GB)
Terminal
Update available
35°C Mostly cloudy
ENG IN
07:02 PM 15-04-2026


docker desktop PERSONAL
Search Ctrl+K
Gordon BETA
Containers
Images
Volumes
Kubernetes
Builds
Models
MCP Toolkit BETA
Docker Hub
Docker Scout
Extensions
Upgrade plan
Images Give feedback
Local My Hub
2.27 GB / 5.67 GB in use 10 images
Last refresh: 19 hours ago
Search
Name Tag Image ID Created Size Actions
kindest/node v1.34.3 08497ee19eac 4 months ago 1.35 GB
docker/desktop-cloud-provider-kind v0.5.0 4ad59ce20658 4 months ago 595.06 MB
docker/desktop-containerd-registry v0.0.3 179a9ea3ae47 4 months ago 50.32 MB
notes-app latest 2c38ae101e70 7 hours ago 768.47 MB
mfa-auth-service 1.0.0 6fff79117398 3 hours ago 716.07 MB
gcr.io/k8s-minikube/kicbase v0.0.50 b97074569ae9 2 months ago 1.93 GB
gcr.io/k8s-minikube/kicbase <none> eb4fec00e8ad 2 months ago 1.93 GB
23mis0634/online-auction-system latest 00ae5ffb1fc2 21 minutes ag 635.95 MB
auction-app latest 69d0eaccd423 47 minutes ag 635.95 MB
auction1-app latest 90c2ec96099c 8 minutes ago 675.08 MB
Showing 10 items
Engine running
Kubernetes running RAM 6.96 GB CPU 2.52% Disk: 11.84 GB used (limit 1006.85 GB)
Terminal
Update available
35°C Mostly cloudy
ENG IN
07:15 PM 15-04-2026

```
PS C:\Users\rakes\Downloads\online-auction-system> docker build -t auction1-app .
[+] Building 2.8s (9/9) FINISHED                                                                                    docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                                                0.0s
 => => transferring dockerfile: 183B                                                                                                0.0s
 => [internal] load metadata for docker.io/library/eclipse-temurin:21-jdk                                                           2.3s
 => [auth] library/eclipse-temurin:pull token for registry-1.docker.io                                                              0.0s
 => [internal] load .dockerignore                                                                                                   0.0s
 => => transferring context: 2B                                                                                                     0.0s
 => [1/3] FROM docker.io/library/eclipse-temurin:21-jdk@sha256:c97d5f4e7e04cc6bfcd9fd2022ab689d181463d7e227c31ce2c84093e6a963a4      0.0s
 => => resolve docker.io/library/eclipse-temurin:21-jdk@sha256:c97d5f4e7e04cc6bfcd9fd2022ab689d181463d7e227c31ce2c84093e6a963a4      0.0s
 => [internal] load build context                                                                                                   0.0s
 => => transferring context: 86B                                                                                                    0.0s
 => CACHED [2/3] WORKDIR /app                                                                                                       0.0s
 => [3/3] COPY target/online-auction-system-1.0.0.jar app.jar                                                                       0.0s
 => exporting to image                                                                                                              0.2s
 => => exporting layers                                                                                                             0.1s
 => => exporting manifest sha256:452aa3fccbf0c79aed8859a425ed61b0037892018718850709541665d743f13d                                  0.0s
 => => exporting config sha256:1e2fd32206208e57d72473112799f7ce6c4b454b3910628f625522e12b900d0e                                     0.0s
 => => exporting attestation manifest sha256:bbe31b9bd79a1be06caabee39c1b4934d44f661abbf10c121443f9825d56df97                       0.0s
 => => exporting manifest list sha256:90c2ec96099ca68d4a382c0bba6b24d0570394cfa814ce03e45147670be60a31                              0.0s
 => => naming to docker.io/library/auction1-app:latest                                                                              0.0s
 => => unpacking to docker.io/library/auction1-app:latest                                                                           0.0s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/iwxzuvhq2he1jbjvtjn7c0uf4
PS C:\Users\rakes\Downloads\online-auction-system> docker run -p 8082:8080 auction1-app
Highest bidder: Priya
Highest bid: 600
PS C:\Users\rakes\Downloads\online-auction-system>
```

Kubernetes:

```
PS C:\Users\rakes\Downloads\online-auction-system> minikube start
😄  minikube v1.38.1 on Microsoft Windows 11 Home Single Language 25H2
✨  Using the docker driver based on existing profile
👍  Starting "minikube" primary control-plane node in "minikube" cluster
🚜  Pulling base image v0.0.50 ...
🔄  Restarting existing docker container for "minikube" ...
🐳  Preparing Kubernetes v1.35.1 on Docker 29.2.1 ...
🔎  Verifying Kubernetes components...
    ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
🌟  Enabled addons: default-storageclass, storage-provisioner
🏄  Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
PS C:\Users\rakes\Downloads\online-auction-system> kubectl apply -f k8s/deployment.yaml
deployment.apps/online-auction-system created
PS C:\Users\rakes\Downloads\online-auction-system> kubectl get pods
NAME                                    READY   STATUS             RESTARTS      AGE
mfa-auth-service-6fb659cc9f-j7djm       1/1     Running            1 (35s ago)   163m
online-auction-system-dc6d86c77-m25hm   0/1     InvalidImageName   0             6s
PS C:\Users\rakes\Downloads\online-auction-system> kubectl get svc
NAME               TYPE        CLUSTER-IP     EXTERNAL-IP   PORT(S)          AGE
kubernetes         ClusterIP   10.96.0.1      <none>        443/TCP          164m
mfa-auth-service   NodePort    10.97.25.46    <none>        8080:30081/TCP   163m
PS C:\Users\rakes\Downloads\online-auction-system> minikube service auction-service
```

```
PS C:\Users\rakes\Downloads\online-auction-system> kubectl get services
NAME                TYPE        CLUSTER-IP    EXTERNAL-IP   PORT(S)          AGE
kubernetes          ClusterIP   10.96.0.1     <none>        443/TCP          167m
mfa-auth-service    NodePort    10.97.25.46   <none>        8080:30081/TCP   166m
PS C:\Users\rakes\Downloads\online-auction-system>
```

Describe what to build

Defau...

Ln 1, Col 24 Spaces: 4 UTF-8 LF { } Dockerfile Finish Setup Iniciar Go Live

Search

ENG IN

07:15 PM
15-04-2026